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SESSION 4: EMERGING CHALLENGES FOR DISCOMs — GW-SCALE AI DATA CENTRES, MW-SCALE EV CHARGING AND GRID-LEVEL POWER QUALITY

Power Quality for India's New Grid

Presented By

ANIL KUMAR, DIRECTOR, ENSPAR

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



Five messages

1. **Every new connection is power-electronic**

   Solar and wind plants, batteries, EV chargers, data centres and electrolysers all connect through converters, not rotating machines.

2. **Each type brings a different PQ problem**

   Oscillations at RE parks, voltage rise and EMI from rooftops and chargers, harmonics from rectifiers, ride-through risk from data centres.

3. **India checks PQ at connection, then mostly stops**

   CEA standards set limits, but only frequency and average power factor are routinely monitored afterwards.

4. **Measurement must follow the connection**

   Class A meters at points of connection, AMI at the edge, one open PQ database feeding utility systems.

5. **Foresee and prevent**

   Many outages can be foreseen and prevented by PQ disturbance monitoring and analytics.

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01

THE NEW GRID

WHO IS CONNECTING?

WHY DISTURBANCE LEVELS ARE RISING?



The numbers behind the change

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71 → 280 GW

variable renewable capacity, 2020 actual to 2030 projected

289 → 459 GW

projected peak demand, 2026–27 to 2035–36 (CEA)

26.3 GW

projected AI data-centre demand by FY32 (MoP), against 1.57 GW installed in Aug 2026

1.3–3.9 mn

public chargers needed by 2030 for 80 million EVs, up from about 29,000 today

7.24 cr

smart meters installed of 20.33 crore sanctioned under RDSS (June 2026)

Almost all of this new capacity, demand, charging and metering connects through power electronics. That changes what the grid has to carry and what it has to measure.

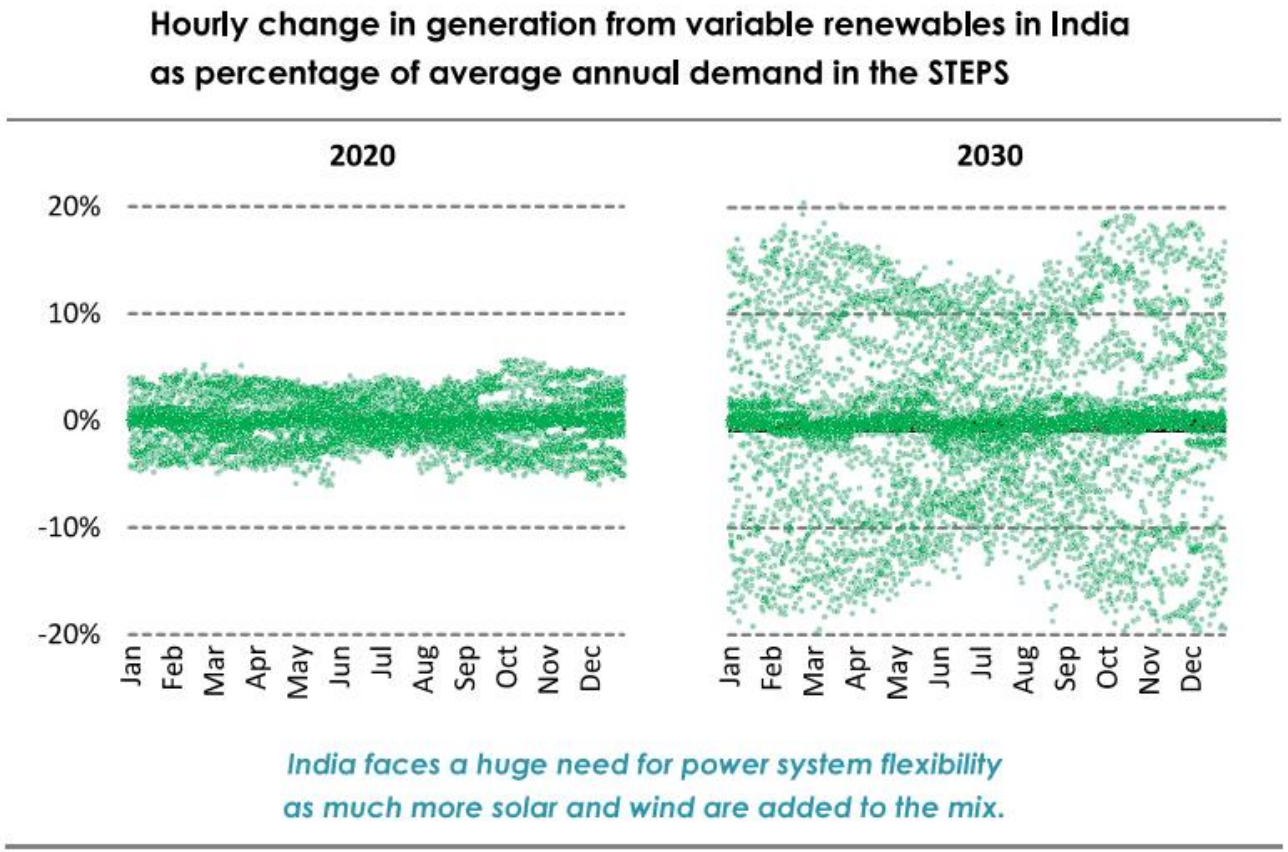
Sources as cited in the source decks; EV figures from CII and ORF reports as compiled in the EV note (electric vehicles by 2030)

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Who is connecting, and at what voltage

Transmission · 220 kV and above	**GW-scale solar and wind parks**	**Large data centres**	**Large industries, BESS, hydrogen**	**Traction: Western DFC (220/2×25 kV)**	***Shared nodes: RE ramps, load steps and trips interact; grid stability at risk***
Sub-transmission · 33–132 kV	**Data-centre and C&I campuses**	**EV bus depots and hubs (5–10 MW)**	**Metros and 132 kV traction**		*Sag sensitivity, harmonics, flicker, unbalance*
Distribution · 11 kV and LT	**Rooftop solar**	**Home, workplace and public chargers**	**LED, AC and IT loads**		*Voltage rise, unbalance, DC injection, EMI, DT overload*

Impact on grid stability: voltage and frequency variation



Inadequate balancing and ramping solutions will impact grid stability.

Variable renewable capacity: 71 GW in 2020, 280 GW projected for 2030.

AT 220 kV AND ABOVE, THE SAME NODES CARRY

GW-scale solar and wind: Ramps and oscillations (Bhadla, Khavda 2026)

Large data centres: Load steps and sudden trips (Virginia 2024, 2026)

Large industries: Dynamic loads cause wide reactive power swings and voltage instability; dips hit sensitive loads

Traction (Western DFC): Single-phase 2×25 kV load: unbalance, harmonics, fast load swings

Chart: hourly change in variable renewable generation as % of average annual demand, India, Stated Policies Scenario (IEA), from Enspar slides

Why disturbance levels are rising

Falling system strength

Inverter-based resources add little short-circuit power, so the same emissions cause more voltage distortion, especially at low-SCR RE pooling stations.

Supraharmonics

Chargers, inverters and rectifiers switch at kilohertz rates, producing emissions from 2 to 150 kHz that classic limits do not cover.

Conducted and radiated EMI

Fast SiC and GaN switching creates high-frequency noise that disturbs meters, PLC communication, protection and control electronics.

Controls that interact

Plant controllers, STATCOMs and inverters can excite each other, causing oscillations from 0.1 Hz to about 20 Hz.

DC offset

DC injected by grid-connected inverters saturates transformer cores, overheats them and corrodes earthing electrodes.

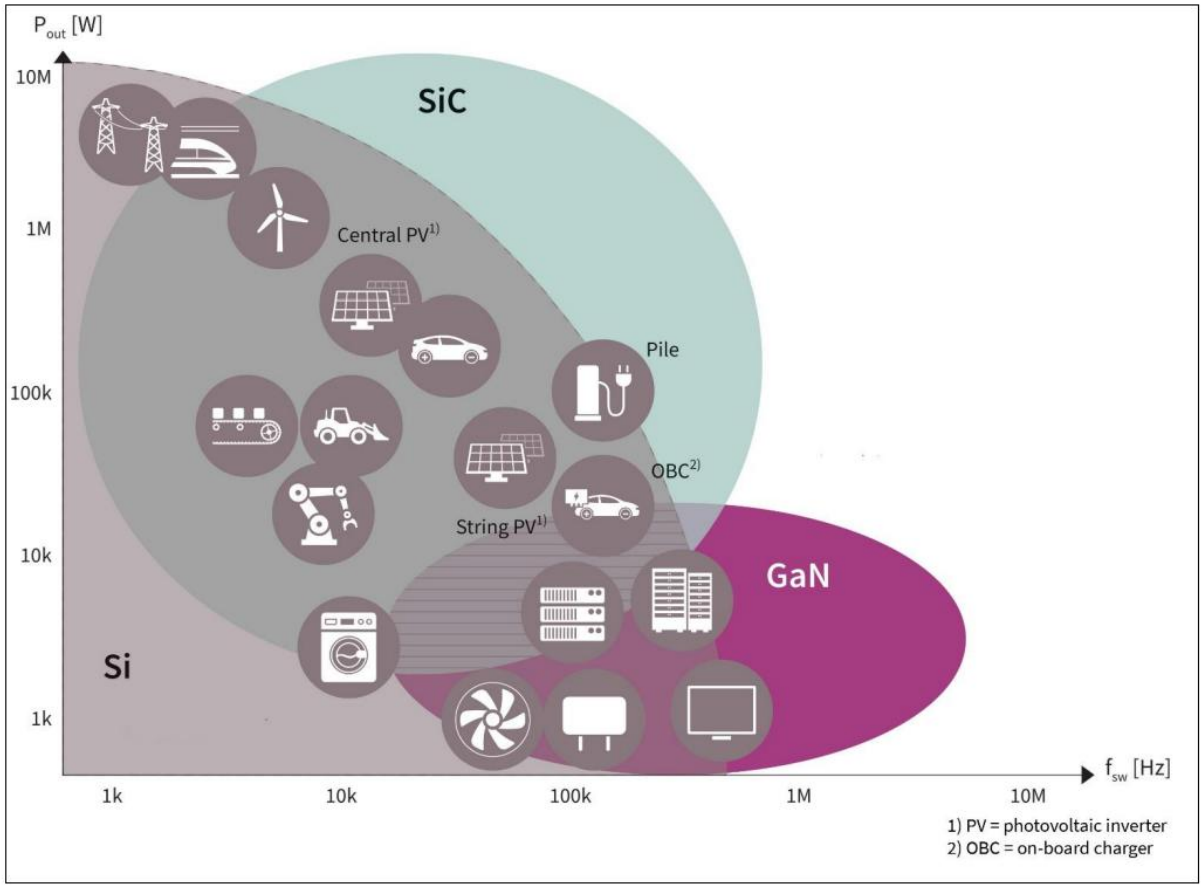
Resonance: a major concern

Cables, power electronics, static transformers and HVDC add capacitance, so the grid loses its inductive character. Resonances above 2 kHz then amplify emissions and raise instability risk.

Proposal from the PQ Intelligence deck: treat grid impedance as a PQ parameter when assessing future grids

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EMI and supraharmonics: what classic PQ monitoring misses



Where the noise comes from

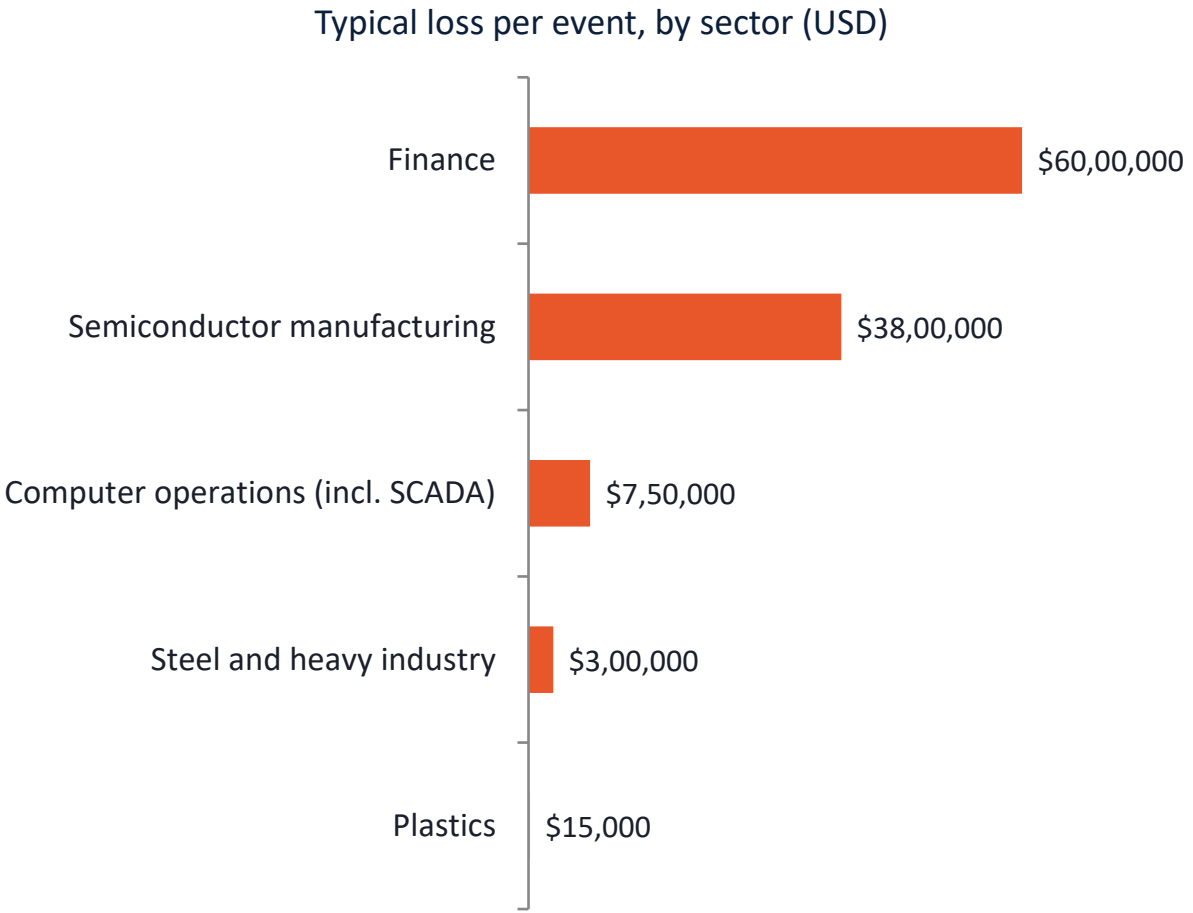
PV inverters switch at about 10–20 kHz, EV chargers at 10–100 kHz and LED drivers up to 30 kHz. Active filters and SVGs switch fast too, and add EMI unless precautions are taken at equipment level. Grid devices are specified only for 50 Hz.

How to measure it

- Analysers that cover harmonics, interharmonics and supraharmonics at the same time
- At the actual grid connection point, on the MV side
- Grid-impedance measurement to find resonances

Chart: Infineon CoolSiC application note. Measurement practice: NEO Messtechnik (PQA8000H at a PV plant, Sep 2026); LV-side readings are distorted by the transformer

The cost of poor power quality: global evidence



UNITED STATES

$145–230 bn	a year in economic losses from power quality (EPRI/DOE)
$80 bn	a year from power interruptions alone (Berkeley Lab, 2005)
$250,000	lost in a day of paper-mill production from one voltage sag (Business Week, 1996)
1 in 2	computer problems, and a third of data loss, trace back to the power line (Contingency Planning Research)

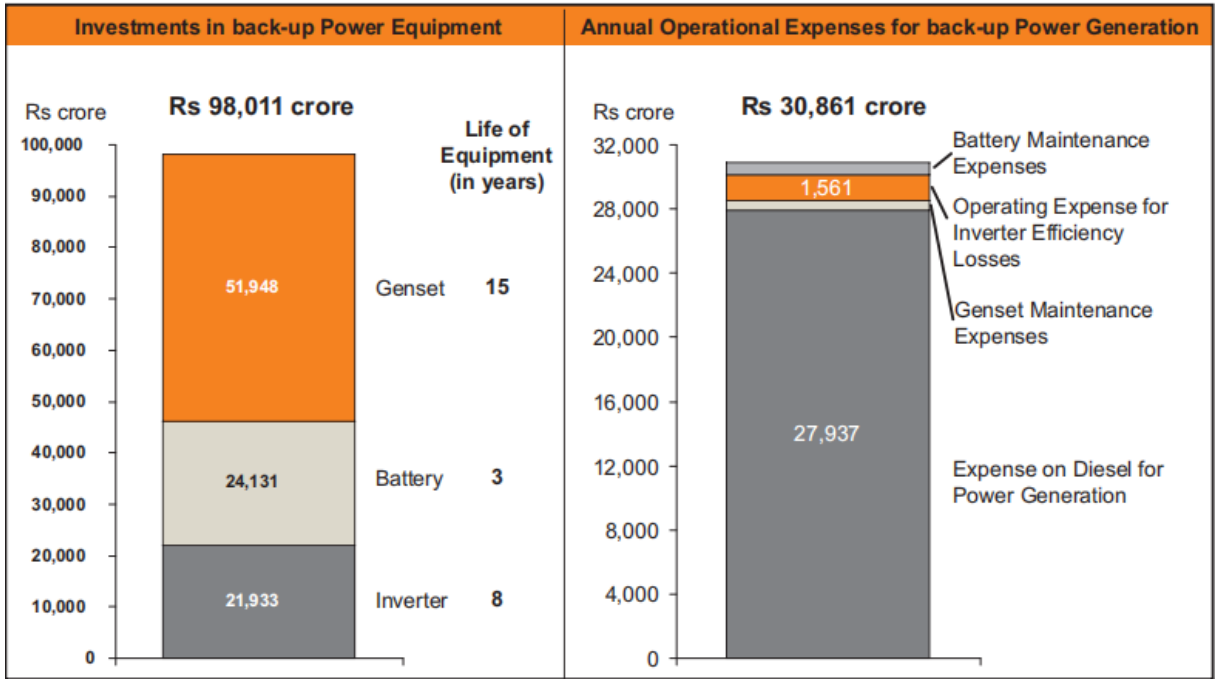
Sector figures as cited in the International Industry Practice on PQ Monitoring deck; US figures as compiled from EPRI/DOE and the sources named

Economic impact of power quality in India: about 5% of GDP

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Power quality = Voltage quality + Availability + Service quality



Backup power: Rs 98,011 crore invested in gensets, batteries and inverters, plus Rs 30,861 crore a year to run them, mostly diesel. It also releases millions of tonnes of CO_2 a year.

~5% of GDP	cost of power outages to Indian industry: backup kit, diesel, unstable work cycles, lost profit
Rs 20,000 cr	a year in direct downtime costs (utilities): 57% sags and short interruptions, 35% transients and surges
$3.5 mn / hour	cost of the October 2020 Mumbai outage; restoration took 8–10 hours
Rs 20.4 bn	a year in downtime from PQ disturbances in the pharma industry alone

Many of these outages can be foreseen and prevented by PQ disturbance monitoring and analytics.

Sources: IJETMAS (March 2015); CIRED (June 2009); Wärtsilä, The Real Cost of Power (21 cities, 1,500 respondents)

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02

POWER QUALITY BY CONNECTION TYPE

WHAT EACH NEW SOURCE AND LOAD BRINGS, AND WHAT THE RULES SAY



Each connection type has its own PQ signature

Connection type	Harmonics	Supra-harmonics (2–150 kHz)	EMI (conducted, radiated)	Voltage rise, fluctuation, flicker	Unbalance	Sag sensitivity, ride-through	DC injection	Oscillations, control interaction
Solar and wind parks	Med	Med	Med	**High**	Low	**High**	Low	**High**
Rooftop solar	Med	Med	**High**	**High**	**High**	Med	**High**	Low
BESS and hybrid plants	Med	Med	Med	Med	Low	**High**	Med	Med
EV charging	**High**	**High**	**High**	Med	**High**	Med	Med	Low
Data centres and load centres	**High**	Med	Med	Med	Low	**High**	Low	Med
Large C&I: furnaces, drives	**High**	Med	Med	**High**	**High**	**High**	Low	Low
Electrolysers and traction	**High**	Med	Med	Med	**High**	Med	Low	Med

Indicative assessment drawn from the source decks, Indian grid events and literature. Site-specific studies decide the actual risk.

Solar and wind parks: weak grids and oscillations

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68 RE-complex disturbances, Jan 2022 – Jun 2025

1–7 GW of generation lost per event, mostly in Rajasthan and Gujarat, linked to LVRT/HVRT non-compliance (CERC). Enforcement has cut their frequency; CERC also urges STATCOMs.

Bhadla–Bikaner–Fatehgarh, 11 May 2026

Sustained oscillations across seven RE projects (3,168 MW). The 400 kV Bhadla bus swung between 373 and 440 kV. PMU data was missing for projects totalling 6.7 GW.

Khavda complex, 13 May 2026

About 9 GW of generation lost, the largest single event in Indian grid history. 0.458 Hz oscillations and LVRT/HVRT behaviour of RE plants were implicated.

OSCILLATION TYPES SEEN BY GRID-INDIA

0.1–0.2 Hz	Voltage oscillations from plant-controller response times
3.5–7 Hz	Linked to STATCOM gain settings
8–20 Hz	Sub-synchronous, when short-circuit ratio drops after outages

Response: CEA has decided to mandate grid-forming inverters for new BESS above 50 MW and for 25% of new RE capacity. PQ role: high-resolution recording at every point of interconnection, alongside PMUs, so root causes can be found in days, not months.

Sources: GRID-India via Mercom (May 2026); Power Peak Digest on NCT 38th meeting, Khavda event and CEA GFM decision (2026); CERC Chairperson via Down To Earth (Aug 2026)

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Rooftop solar: voltage rise and EMI at the grid edge

Voltage rise	Midday export lifts LT voltage above limits, tripping inverters and damaging appliances
Phase unbalance	1–10 kW single-phase systems load phases unevenly; neutral current grows
Reverse power flow	Protection and tap settings were designed for one-way flow
DC injection	Limit of 0.5% of rated output under CEA standards for prosumers
EMI and supraharmonics	Inverter switching at 10–20 kHz interferes with smart-meter PLC and nearby electronics
Mass tripping	Thousands of inverters reacting to one disturbance act like one large plant

What works

- Smart inverter settings (volt-var, volt-watt) to IS 18968:2025
- EMC-tested inverters with adequate output filters
- Active filters only with EMC precautions at equipment level, or they add EMI
- Hosting-capacity maps by DT from measured voltage
- Phase balancing of new connections
- DT monitoring and OLTC on solar-rich feeders
- Community or DT-level BESS to absorb midday surplus

Battery storage and hybrid plants: problem and solution

As a connection

- Converter harmonics and supraharmonics
- Transients at switching, islanding and reconnection
- Ride-through behaviour must match grid code
- Control interaction with nearby RE plants and STATCOMs
- Draft CEA 2026 regulations: dedicated ESS standards; black start for BESS of 50 MW and above

As a solution

- Grid-forming BESS adds system strength and damps oscillations
- Fast voltage and reactive support at weak feeders
- Peak shaving: midday solar into the evening peak
- Relief for overloaded DTs at EV hubs
- Ride-through support for sensitive industrial loads

CEA has decided to mandate grid-forming inverters for all new BESS above 50 MW (2026)

EV charging and V2G: harmonics, EMI and transformer stress

Harmonics	Charger rectifiers draw distorted current; fast-charging hubs dominate the local spectrum
Transformer derating	Harmonic losses heat DTs; IEEE C57.110 derating needed at bus and fleet depots
Evening peak	Uncontrolled charging stacks on the residential peak and overloads DTs
Unbalance	Two- and three-wheeler and home chargers are single-phase
Supraharmonics, EMI	Fast switching emits 10–100 kHz noise into the grid and nearby electronics
V2G	Bidirectional chargers add reverse flow, islanding risk and new emissions; no Indian V2G standard yet

Rules: CEA below-33 kV standards apply IEEE 519 to charging stations. Tamil Nadu's Supply Code (2022) mandates IEEE 519 compliance for charging stations and consumers at 11 kV and above.

What works

- Active front-end chargers and filters, with EMC precautions
- Smart charging in solar hours
- DT monitoring before and after
- Co-located BESS at hubs

Industry, electrolysers and traction: the heavy rectifier loads

Large C&I

- Induction and arc furnaces: harmonics and flicker
- Variable-speed drives: harmonics
- Dynamic loads: wide reactive power swings, voltage instability
- Motor starting: sags
- Biggest source of complaints and penalties today

Green hydrogen electrolysers

- MW-scale rectifiers: 5th, 7th, 11th, 13th harmonics
- Large reactive power demand with thyristor designs
- IGBT rectifiers cut distortion at higher cost
- e.g. 4 MW rectifier blocks for IOCL's 10 kTPA project

Railway traction and metros

- 25 kV single-phase traction draws unbalanced current
- Harmonics from locomotive converters
- Regenerative braking returns power
- Western DFC substations draw from 220 kV (220/2×25 kV)

GRID-India has proposed extending CEA's solar and wind standards to inverter-based storage and electrolysers.

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India’s PQ rulebook by connection

Regulation	Applies to	PQ requirements
CEA Connectivity to the Grid Regulations 2007, amended 2013 and 2019	Generators incl. solar and wind parks	Harmonics (IEEE 519), flicker, DC injection, reactive power, ride-through
CEA Connectivity below 33 kV (2013, amended 2019)	Rooftop solar, prosumers, EV charging stations	IEEE 519 at PCC; DC below 0.5%; sags, swells, flicker per BIS/IEC
Draft CEA Connectivity Regulations 2026 (replaces 2007)	Generators, ESS, HVDC, FACTS, bulk consumers at 33 kV and above	IEEE 519 and IEEE 2800; flicker per IEC 61000-3-7; unbalance 3%; step change 1.5%; DC below 0.5%; annual PQ measurement
CEA decision on grid-forming inverters (2026)	New BESS above 50 MW; 25% of new RE	System strength and oscillation damping
State supply codes, e.g. TNERC 2022	Consumers at 11 kV and above; charging stations	IEEE 519 compliance, with penalties for excess injection

The gap: limits are checked at connection. Ongoing compliance is rarely measured once the plant or load is running.

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03

SPOTLIGHT: DATA CENTRES AND EV CHARGING

GW-SCALE AI DATA CENTRES AND MW-SCALE CHARGING: THE LOADS DISCOMS MUST PLAN FOR NOW



India’s data-centre load is arriving fast

1,575 MW

installed data-centre capacity (Aug 2026), up from 375 MW in 2020

17 GW

CEA’s expected data-centre demand by 2031–32

26.3 GW

Ministry of Power projection for AI data centres by 2031–32

CEA committee (Oct 2026)

Set up to assess data centres’ impact on the power system and grid stability

GRID-India’s warning

Spiky data-centre loads and sharp ramps could pose risks to grid stability

Concentrated hubs

Mumbai, Chennai, Hyderabad, Noida and Vizag; AP’s deemed licence for a 1 GW Vizag campus

Sources: KNN India (Oct 2026); Energetica India (July 2026); AI Power India deck. Estimates vary widely, which is itself a planning risk.

India’s EV charging load is arriving fast

80 million

electric vehicles targeted by 2030

29,000 → 1.3–3.9 mn

public chargers today vs needed by 2030

1 : 235

public chargers per EV (July 2025), against 1 : 6–20 globally and a 1 : 40 target

Charger mix

Home and workplace AC 70–80%, public AC 10–15%, DC fast 10–15%; 400,000+ new chargers a year

MW-scale hubs

Public stations with hundreds of DC fast chargers; bus depots needing 5–10 MW in space-constrained cities

Spatial targets

A station every 3 km in cities, every 25 km on highways, DC fast every 100 km for heavy vehicles

Downtime

18–50% of public chargers face high downtime, partly from grid voltage issues

V2G

Bidirectional charging: ISGF AC V2G demo (Delhi DISCOMs, ANERT Kerala); KSEB–IIT Bombay field pilot

Sources: CII and ORF reports, The Hindu, MMCM, as compiled in the EV note; MW-scale hub figures from the DUM Session 4 note; V2G: ISGF (May 2026), IIT Bombay and KSEB

Inside a data centre: where the PQ issues come from

Grid supply	Transformers	UPS and batteries	IT load
Dual feeds, N-1, 33–220 kV	Dedicated substation and switchgear	Double-conversion UPS	PDUs, server supplies, GPUs
Upstream sags and swells arrive here first	Harmonic heating; derating or K-rated units	Rectifier harmonics; transfer to battery on shallow dips	Can run at leading power factor at light load; AI training swings
Cooling		**Backup and storage**	
Chillers, pumps and fans on VFDs	Drive harmonics; motor-starting sags	Diesel sets and on-site BESS	Transfer away and back shows up as sudden load steps

Net effect at the point of connection: a large power-electronic load that is sensitive to the grid and can disturb it in return.

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A two-way risk: the grid hits the data centre, and back

Grid → data centre

- Sags from routine faults trigger UPS transfers
- Swells and transients stress power supplies
- Harmonic voltage from neighbours adds heating
- Interruptions cost about $10,000 a minute for server operations
- Siting decisions rest on anecdote, not measured supply quality

Data centre → grid

- Hundreds of MW can vanish at once when UPS systems transfer
- Sudden load loss pushes voltage and frequency up
- AI training ramps hundreds of MW within seconds
- Harmonics and reactive power at the supply point
- A rapid return after a disturbance hits the grid as a load surge

Power quality and power availability have to be engineered together, at the design stage rather than retrofitted.

When large loads trip: lessons from Virginia

Lesson from Virginia, USA

In July 2024 a routine 230 kV fault made about 1,500 MW of data-centre load switch to backup power at once, across 60 load points. PJM reported nearly 4,000 MW doing the same on 22 July 2026. Sudden load loss pushes voltage and frequency up.

Sag sensitivity	UPS systems transfer on shallow dips, so the grid sees a block of load vanish
Harmonics	Server power supplies, UPS and cooling drives
Load swings	AI training runs ramp hundreds of MW within seconds

What India can do now

- Draft CEA 2026 regulations extend compliance to inverter-interfaced bulk consumers of 50 MW and above
- Add ride-through and ramp-rate rules for large loads
- Class A monitoring at every campus supply point
- Model large loads in connection studies

Sources: NERC Incident Review, Large Load Loss (Jan 2025); PJM via American Public Power Association (2026); Power Peak Digest on draft CEA regulations (2026)

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A connection code for data centres

Requirement	What it means	Status in India
Fault ride-through	Stay connected through defined dips; no mass UPS transfer	Draft CEA 2026, for inverter-based bulk consumers
Harmonic limits	IEEE 519 at the point of common coupling	Existing CEA standards; draft CEA 2026
Power factor and reactive power	Stay inside a band, including leading operation	Draft CEA 2026
Annual PQ measurement	Measured compliance, not a one-time test	Draft CEA 2026
Validated dynamic models	Models for studies, revalidated every five years	Draft CEA 2026
Ramp-rate limits and staged return	Cap MW-per-minute swings; reconnect in steps	Not yet specified; under development in the US
Continuous Class A monitoring	PoC data shared with the SLDC for event analysis	Proposed in this deck
UPS and BESS as grid assets	Fast frequency response and demand response	Proposed in this deck

Draft CEA (Technical Standards for Connectivity to the Grid) Regulations 2026, as reported by Solar Quarter and Power Peak Digest; check the final notified text

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04

MEASURING POWER QUALITY

WHERE INDIA STANDS AND HOW TO BUILD THE MONITORING LAYER



Regulated on paper, barely measured in practice

Monitored today

- Frequency
- Average power factor

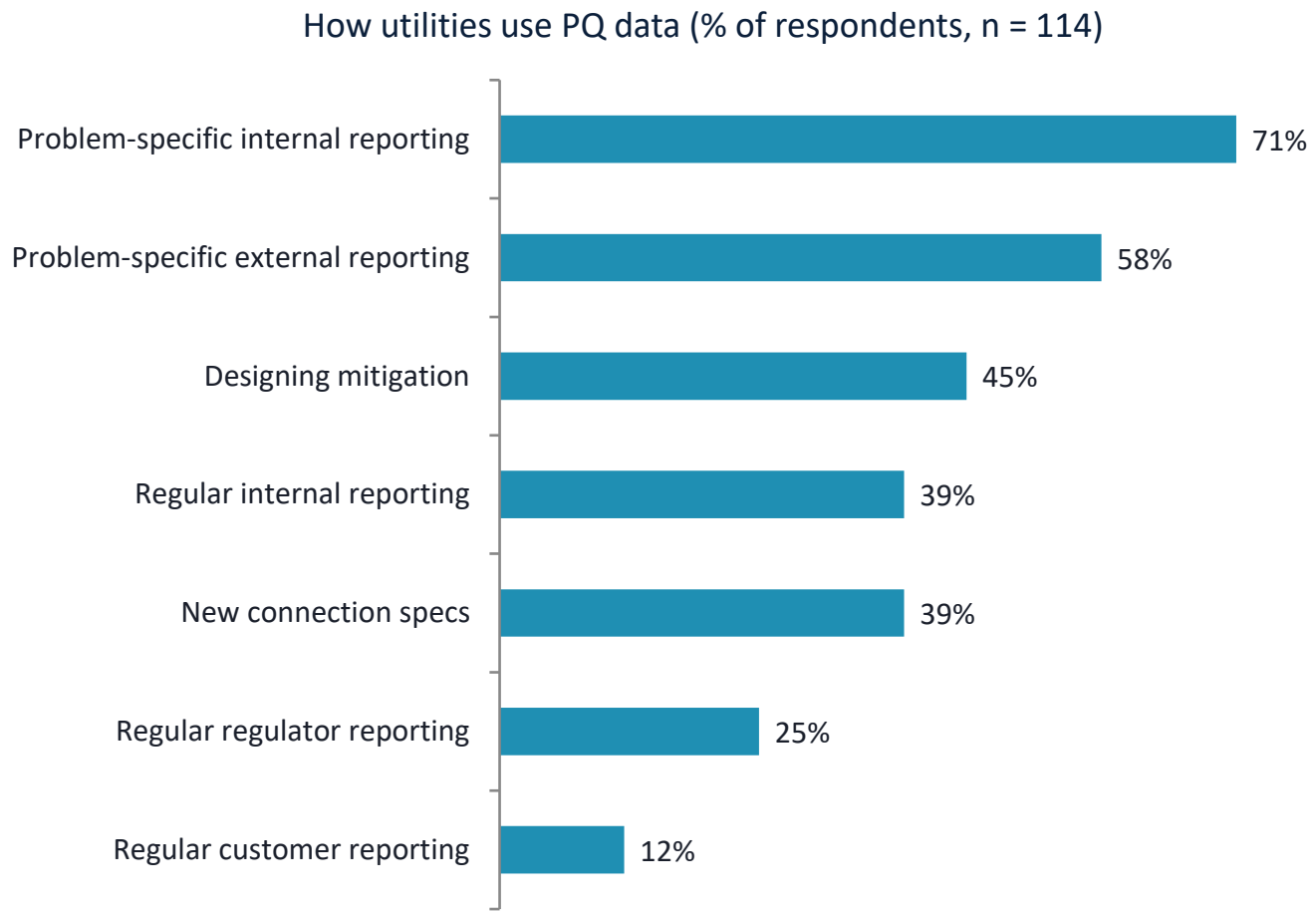
Specified, not strictly monitored

- Voltage variation at supply points
- Harmonics (CEA limits)
- Voltage unbalance

State	Provision	Status
Tamil Nadu	Consumer pays 15% of tariff for harmonic injection above limits	Defined
Maharashtra	Consumer must prevent injection; penalty not yet specified	Partial
Andhra Pradesh	DISCOMs to monitor voltage and harmonics; no recording method	Partial
MP, Gujarat, Karnataka, Delhi	No clear monitoring method or penalty	Not defined

Private utilities lead: Tata Power, Adani Electricity and BSES run substation PQ meters with central databases. Public tenders are built on AMR/AMI specs and need rewriting.

Global practice is still reactive: India can leapfrog



93%

monitor mainly to answer customer complaints

39%

use PQ data to write new connection specifications

25% · 12%

report regularly to regulators · to customers

Source: IEEE Transactions survey as presented in International Industry Practice on PQ Monitoring

Smart meters alone cannot carry power-quality data

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Smart meters (IS 16444)

- Every consumer: dense coverage
- Per-phase voltage profile, interruptions
- Over/under-voltage, PF
- Some sags, flicker, THD, with limited accuracy
- About 10 MB of PQ data per meter per year

~730×

more data per year from one PQ meter than a smart meter's PQ bandwidth

20 MB × 365 ÷ 10 MB

Class A PQ meters (IEC 61000-4-30)

- At points of connection: sparse but precise
- Waveform-level event capture
- Full harmonic spectrum, flicker, dips, transients
- About 20 MB per meter per day
- Too much for today's HES and MDMS

Design answer: a hybrid architecture. AMI for breadth, dedicated PQ meters for depth, one central PQ database with open APIs to MDMS.

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Monitoring plan: measure at every new point of connection

Connection	Where to measure	Instrument	Priority parameters
Solar, wind, hybrid parks	Point of interconnection, pooling station	Class A meter plus PMU and disturbance recorder	Oscillations, harmonics, supraharmonics, flicker, ride-through
BESS and electrolysers	Point of connection	Class A meter	Harmonics, reactive power, transients
Data centres, large C&I, traction	Supply substation, consumer metering point	Class A meter	Sags, harmonics, flicker, unbalance, load steps
EV hubs and depots	Hub transformer, MV side	Class A meter with supraharmonic range	Harmonics, supraharmonics and EMI, DT loading
Rooftop solar and LT loads	DT, consumer premises	DT monitor; smart meter (IS 16444)	Voltage band, unbalance, reverse flow, outages

IEEE 1159 event categories | **IEEE 1159.3 PQDIF exchange** | **IEC 61850 logical nodes** | **IEC 61000-4-30 measurement**

An open PQ layer that feeds every core system

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DATA SOURCES

- Class A meters at connections
- Smart meters (AMI/MDMS)
- DT monitoring units
- Relays, DRs and PMUs
- SCADA / RT-DAS
- RE, BESS and EV telemetry



Open, device-agnostic PQMS

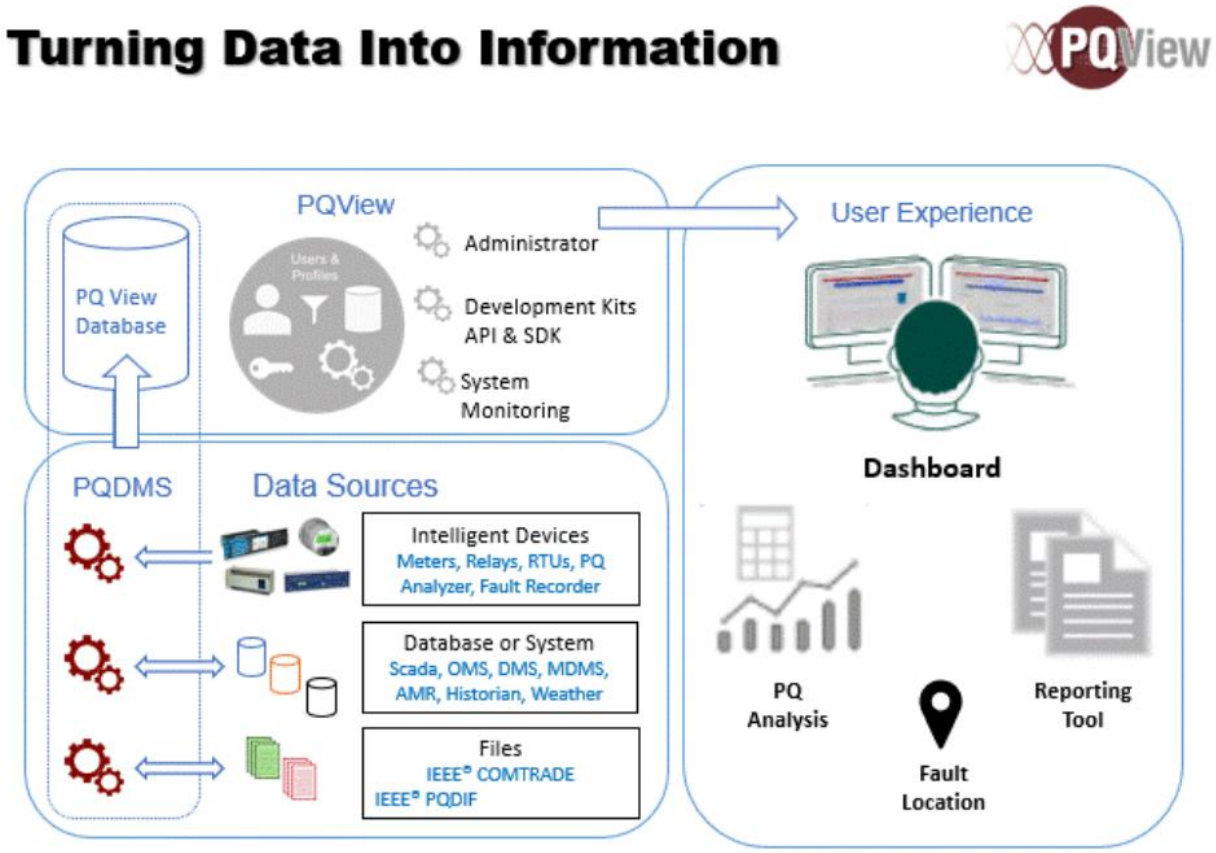
- One event database across all devices
- PQDIF, COMTRADE, IEC 61850 formats
- Compliance: IEEE 519, IEEE 2800, EN 50160
- Waveform signatures and fault location
- Utility-owned data, open APIs
- Cyber-secured, VAPT-compliant

SYSTEMS FED

- ADMS: Volt/VAR, state estimation
- OMS: fault location, FLISR
- SLDC/RLDC: event analysis
- Asset management: health
- DERMS: hosting capacity
- **IES: open PQ data**

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Turning data into information: PQView as an example



Collects from every source

Meters, relays, RTUs, PQ analysers and fault recorders; SCADA, OMS, DMS, MDMS, AMR, historian and weather; COMTRADE and PQDIF files

One database

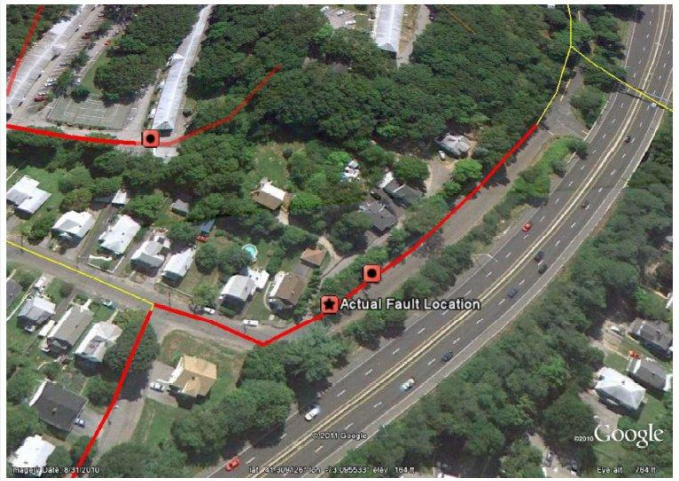
Device-agnostic data management, with administration, API and SDK

Turns it into answers

Dashboards, PQ analysis, reporting and fault location

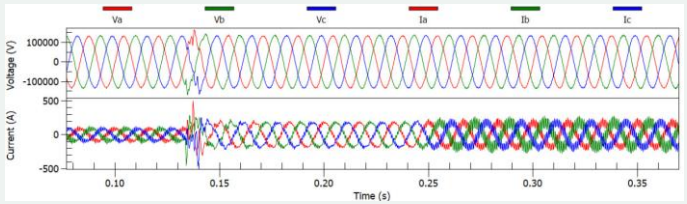
PQView (Electrotek) is used by 74 utilities and operators worldwide, none yet in India. Shown as an example of an open, device-agnostic PQ platform.

Field evidence: waveforms see trouble before it lands



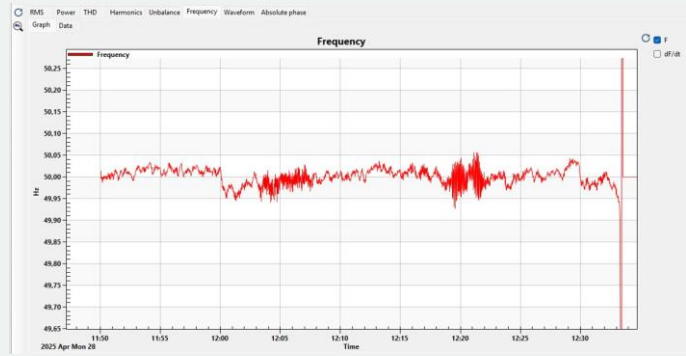
Fault location on the map

Fault distance estimated from recorded waveforms and plotted on GIS. Crews go straight to the fault.



Waveform signatures

Harmonic resonance, breaker restrike, failing tap changers and incipient cable faults leave recognisable signatures.



Iberian blackout, 28 Apr 2025

A PQ meter recorded 0.1–1 Hz frequency oscillations well before the collapse. A trigger could have flagged it.

The Iberian and Indian RE-park events share a pattern: low-frequency oscillations visible in high-resolution data before trouble. Images: PQ Intelligence deck

Waveforms as a cyber sensor: telling an attack from a fault

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What an intrusion leaves in the waveform

- **Switching out of sequence:** a tie breaker closed before the fault cleared, or a breaker opened with no fault current before it
- **Tampered inverter settings:** unusual harmonics or reactive swings with no weather or grid cause
- **Spoofed telemetry:** SCADA reports normal while an independent meter sees the change
- **Hidden reconfiguration:** impedance shifts from upstream switching, even when SCADA shows nothing

How PQ monitoring helps detect it

- **A second, independent data path:** Class A meters and µPMUs, secured separately from SCADA, as a physical reality check
- **Event classification: label each event as a fault, equipment failure or suspected intrusion**
- **Correlation:** match waveform anomalies with OT network logs in the utility's security operations centre and CSIRT-Power alerts
- **Forensics:** time-stamped waveforms show what really happened after an incident

A cyberattack on the grid ends as a physical event, so it shows in the waveform even when telemetry is spoofed. Methods are still at research and pilot stage. Sources: LBNL µPMU cyber-detection project (US DOE); ASU/LBNL (NASPI); DARPA RADICS via IEEE Spectrum

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05

ROADMAP AND RECOMMENDATIONS

CHECK AT CONNECTION, THEN KEEP MEASURING



PQ checks across the connection lifecycle



Study

Harmonic, SCR and hosting-capacity studies; large-load models



Certify

Equipment credentials: IEEE 519, IEEE 2800, IS 18968



Commission

Class A baseline at the point of connection before trial operation



Operate

Continuous monitoring into one PQ database; event analysis



Enforce

Benchmarks, penalties or compensation; disconnection as last resort

The draft CEA 2026 regulations detail studies, equipment data and commissioning, and add annual PQ measurement and five-yearly model validation. Continuous monitoring and enforcement, the orange stages, remain the weakest link.

One roadmap: digitalisation, Energy Stack and power quality

	2026–29 · Foundation	2029–32 · Integrated operation	2032–35 · Intelligence	2035+ · Selective advanced
Digitalisation (CEA)	GIS and master data; AMI, HES, MDMS; SCADA/RT-DAS	GIS–AMI–SCADA–OMS linked; common network model; ADMS pilots	ADMS scale-up; FLISR and Volt/VAR; DERMS, EV charging	Digital twins; DER orchestration; autonomy in safety envelopes
Energy Stack	IES 12-month PoC through the UIP; IES blueprint	National IES rollout; PQ fields in Energy Digest	PQ services market and credentials at scale	Flexibility services on IES rails
Power quality	Class A at new RE, BESS and bulk-load connections; PQ in readiness assessment; open PQMS pilot	PQMS linked to SLDC, GIS, OMS; emission credentials at connection; compensation	PQ inputs to VVO; waveform signatures in asset management; SERC reporting	Real-time PQ automation; oscillation and blackout early warning

Phases from CEA's Unified Framework

Pilots across the connection types

RE pooling station

One low-SCR complex: Class A plus PMUs at every point of interconnection

Solar-heavy DISCOM

Rooftop-rich feeders, large C&I and EV hubs; DT monitoring

BESS DISCOM

Distribution BESS measured before and after; PQ services trial

Data centre and EV hub

One data-centre campus and one MW-scale bus depot; ride-through, harmonics, EMI

Measure of success	What it shows
Share of new connections with Class A data in the central PQ database	Is monitoring following the connection?
Oscillation and trip events with root cause found within 15 days	Is the data good enough to act on?
Meter vs reference-analyser agreement; consumers within voltage band	Accurate enough for compensation? Baseline quality
Hosting capacity published vs DER and EV applications approved	Does measured headroom change decisions?

Risks and mitigations

Risk	Mitigation
Developers treat PQ as a one-time connection test	Continuous monitoring as a condition of connection for large plants and loads
High-resolution data missing when events happen (PMU data was 71% available at Bhadla)	Data recording, retention and sharing written into connection agreements
DISCOMs see transparency as penalty exposure	Measurement-only period first; benchmarking before enforcement
PQ meter tenders copied from AMR/AMI specs	Open, device-agnostic PQMS spec with waveform capture and utility-owned data
Schema fragments across states	National core via Forum of Regulators; state extensions on top
New devices widen the attack surface	Treat as OT assets under CEA cyber regulations and IEC 62443 from day one

Seven recommendations

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1 — Monitor after connection, not only at it

Continuous Class A monitoring for RE parks, BESS, bulk loads and EV hubs

2 — Finalise the 2026 connectivity rules with data duties

Recording, retention and sharing of PQ and PMU data

3 — Set ride-through rules for large loads

Data centres and electrolysers, with ramp-rate limits

4 — Build the hybrid measurement architecture

AMI for breadth; Class A at connections; one open PQ database

5 — Make PQ a DISCOM and STU KPI

Report voltage, harmonic and unbalance indices to SERCs

6 — Share PQ data openly

Publish PQ indices to SERCs and through the India Energy Stack

7 — Run pilots across connection types

RE complex, two DISCOMs, a data centre and an EV hub, from April 2027

The immediate ask: data duties in the final 2026 connectivity regulations, and committed pilot sites.

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KNOW WHERE WE ARE

“If we could first know where we are, and whither we are tending, we could better judge what to do, and how to do it.”

Abraham Lincoln, 16 June 1858

Every new connection should arrive with a way to measure what it does to the grid.



Appendix: sources added in this version

Topic	Source
Draft CEA Connectivity Regulations 2026	Solar Quarter (July and Aug 2026); Power Peak Digest (2026)
CEA below-33 kV standards: IEEE 519, DC injection	CEA draft amendment to the 2013 regulations; Mercom
Tamil Nadu supply code, IEEE 519 for 11 kV+ and charging stations	TNERC press release (June 2024)
Bhadla oscillation, 11 May 2026	Mercom, citing GRID-India; Energetica India
Oscillation types at RE complexes	Power Peak Digest on NCT 38th meeting
Khavda event and CEA grid-forming decision	Power Peak Digest (2026)
Electrolysers and RE event report	Power Line on GRID-India report; Renewable Watch (July 2026)
Data-centre load loss	NERC Incident Review (Jan 2025); American Public Power Association; Utility Dive
Data centres in India, draft CEA compliance duties	KNN India (Oct 2026); Energetica India (July 2026); Solar Quarter (Aug 2026)
EV charging targets and gaps	CII, ORF, The Hindu, MMCM (via the EV note); DUM Session 4 note
Grid stability chart; switching frequencies; PQView	IEA (via Enspar slides); Infineon; Electrotek
EMI and supraharmonic measurement	NEO Messtechnik (Sep 2026)
Cost of outages and PQ disturbances	IJETMAS (2015); CIRED (2009); Wärtsilä; EPRI/DOE; Berkeley Lab (2005)
Western DFC traction supply; V2G pilots	DFCCIL tender documents; ISGF press release (May 2026); IIT Bombay; KNN (Oct 2026)

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Thank you

Presented by Anil Kumar · October 2026

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